

Swiss Holter-Monitoring Eligibility Questionnaire – Eligibility-First

This specification defines a multistep, interactive form for assessing a patient's eligibility for a Holter monitor study in Switzerland. The form prioritises collecting contact details, checking contraindications and symptoms, confirming insurance status and providing an early eligibility message. Detailed medical questions are deferred until after eligibility is established. The design aligns with the existing site (<https://smartholter.netlify.app/>) using **shadcn/ui** components and the project's Tailwind theme ¹.

Overview of Steps

- 1. Step 0 – Contact & Account**
Capture email, phone and date of birth with OTP verification.
- 2. Step 1 – Eligibility Gate**
Ask about contraindications, symptom presence, optional family history and insurance status.
Determine early eligibility and self-pay vs insured path.
- 3. Stage 2 – Detailed Information**
Collect detailed symptom and medical history for all continuing users.
- 4. Stage 3 – Review & Consents (insured)**
For insured users, confirm insurance model, collect GP details and prepare referral packet instructions.
- 5. Self-Pay Stage 3 – Payment & Consents**
For self-pay users, collect payment and final consents.
- 6. Completion**
Present next steps based on path (e.g., referral packet download, partner GP booking, kit shipment).

Progress Indicator

Use the shadcn/ui `Stepper` component to display five steps ("Contact", "Eligibility Gate", "Details", "Review/Consents", "Complete"). A "Save & continue later" link should appear after Step 0; clicking it sends a resume link to the verified email address.

Step 0 – Contact & Account

Field	Component (shadcn/ui)	Required?	Validation	Notes
Email address	<code>Input</code> with OTP verification	Yes	Valid email format; send OTP; require correct OTP before proceeding	Used to create an account and send resume link.

Field	Component (shadcn/ui)	Required?	Validation	Notes
Mobile phone number	with OTP verification	Yes	Valid Swiss/EU phone format; send SMS OTP; require correct OTP	Used for two-factor authentication and SMS notifications.
Date of birth	(date picker)	Yes	Must be ≥ 18 years; calculate age on blur	Ensures patient is an adult.

Upon successful verification, store a partial record in the database and enable a “Save & continue later” option.

Step 1 – Eligibility Gate

1.1 Contraindications

Display a group of three yes/no radio questions. Use ☐ for each item. If **Yes** to any, show a red component with the message “Based on your answers, you may not be eligible for a Holter monitor. Please consult your GP or call emergency services.” and prevent progression. The user can exit or return to edit.

Question	Options	Required?	Notes
Are you currently pregnant?	Yes / No	Yes	Pregnancy is not an absolute contraindication but requires caution; flagged for review.
Do you have a pacemaker or implantable cardioverter-defibrillator (ICD)?	Yes / No	Yes	Flagged for cardiologist review.
Have you been hospitalised for a cardiac condition in the past 30 days?	Yes / No	Yes	Flagged for review.

1.2 Symptom Presence

Use a ☐ list inside a titled “Symptoms”. At least one symptom indicates presence; leaving all unchecked indicates no symptoms.

Symptom option	Notes
Palpitations or racing heart	
Dizziness or light-headedness	

Symptom option	Notes
Fainting or loss of consciousness	If selected, a <code>Dialog</code> appears asking: "Did you lose consciousness and sustain injuries or require medical attention?" (Yes/No). If Yes, show emergency advice and require acknowledgement before continuing.
Chest pain or discomfort	If selected, a <code>Dialog</code> asks: "Is your chest pain severe ($\geq 7/10$) or occurring at rest?" (Yes/No). If Yes, show emergency advice.
Shortness of breath	
Other (free text)	Show a <code>Textarea</code> when selected.
None of the above	Checkable. If selected, disable other checkboxes.

1.3 Family History (optional)

Use a `Switch` or `RadioGroup` with Yes/No.

Question: "Has a close relative (parent or sibling) experienced sudden cardiac death or been diagnosed with an arrhythmia?"

This field is optional and does not block progress.

1.4 Insurance Status

Ask: "**Do you have health insurance coverage in Switzerland?**" Use a `RadioGroup` with **Yes** and **No** options. If **Yes**, reveal a `Select` field to choose **insurance model** (Standard/Flex, HMO/Hausarzt, Telmed). If **No**, set `insuranceModel = 'SelfPay'` and skip the model selection.

After the user selects insurance model status, apply the eligibility logic:

1. **Check symptoms:** If the user selected at least one symptom (excluding "None") → proceed. Else, display a neutral `Alert` explaining that screening without symptoms is typically not reimbursed and ask whether they wish to proceed as self-pay. Use Yes/No buttons; if No → finish; if Yes → set `insuranceModel = 'SelfPay'` and continue.
2. **Check contraindications:** Already handled. Contraindications stop progression.

Users who pass this gate proceed to Stage 2. Store the answers for reference and show a green `Alert` ("You may be eligible for reimbursement.") when the user is insured and has ≥ 1 symptom.

Stage 2 – Detailed Symptom and Medical History

This stage is presented to users who (a) have insurance and ≥ 1 symptom or (b) choose to self-pay (insured with no symptoms or uninsured). Use collapsible sections to avoid overwhelming users.

2.1 Symptom Details (optional but recommended)

Field	Component	Required?	Notes
When did your symptoms begin?	(date)	Optional	Show if at least one symptom selected.
How frequently do you experience symptoms?	(Daily, Weekly, Monthly, Sporadic)	Optional	
How severe are your symptoms?	(1-10)	Optional	Label extremes (1 = minimal, 10 = unbearable).

2.2 Prior Arrhythmia or Procedure

Field	Component	Required?	Notes
Have you been diagnosed with a heart rhythm problem or had a recent cardiac procedure?	(Yes/No)	Optional	If Yes, show a for a brief description and date.

2.3 Medications

Field	Component	Required?	Notes
Please list any cardiac medications you are currently taking.		Optional	Encourage users to provide medication names and doses if available.

2.4 File Uploads

Field	Component	Required?	Notes
Upload previous ECGs or referral letters (optional).		Optional	Accept up to 10 MB; file types: PDF, JPG, PNG.

After completing Stage 2, route the user to Stage 3 based on . For self-pay users, skip Stage 3 (insured) and proceed to Self-Pay Stage 3.

Stage 3 – Review & Consents (Insured)

Only users with insurance (Standard/Flex, HMO/Hausarzt, Telmed) reach this stage. Present a summary of their answers, allow them to edit previous steps, and collect final details specific to their model.

3.1 Insurance Model Confirmation

Reconfirm the model selected. Provide a short description of each model:

- **Standard/Flex:** Choose between your **own GP** or a **partner GP** (Medgate).
- **HMO/Hausarzt:** You must use your assigned GP; partner GPs are not allowed.
- **Telmed:** You are required to call a triage hotline (provided by your insurer) before obtaining specialist care.

3.2 GP & Referral Details

For Standard/Flex: * **Own GP:** Collect **GP name, practice name, HIN email, and phone number** using components. On submission, provide the user with a downloadable referral packet (PDF) containing a cover letter, information sheet and pre-filled referral form. Instruct them to deliver this packet to their GP. * **Partner GP:** Display a linking to the Medgate booking portal. Once clicked, record that the referral will be handled via teleconsultation.

For HMO/Hausarzt: * Collect the **assigned GP** details (as above). Provide the referral packet upon submission. Remind users that partner GPs are not permitted.

For Telmed: * Provide a with hotline details. The Telmed clinician will perform triage and issue a referral if appropriate. No GP details are collected here.

3.3 Consents

Display a list of required checkboxes using ☐ components: * **Truthfulness declaration:** "I confirm that the information provided is accurate and complete." * **Emergency acknowledgement:** "I understand that Holter monitoring does not replace emergency medical evaluation. If I develop severe symptoms, I will seek immediate care." * **Telemedicine & data processing consent:** "I agree to telemedicine services and the privacy policy." Provide links.

After all checkboxes are selected, show a button to finalise the submission. On success, display a confirmation screen with next steps:

- **Standard/Flex (own GP):** "Download your referral packet and deliver it to your GP. We will ship your Holter kit once we receive the signed referral via HIN or fax."
- **Standard/Flex (partner GP):** "Your teleconsultation will generate a referral. You will receive an email once your kit is ready to ship."
- **HMO/Hausarzt:** "Download your referral packet and deliver it to your GP. Partner GPs are not available for this insurance model."
- **Telmed:** "Call the Telmed hotline at [number] to complete triage. Follow their instructions to obtain a referral."

Self-Pay Stage 3 – Payment & Consents

Users who are uninsured or who choose self-pay reach this stage. Present a summary of their answers and collect payment using the integrated payment component (e.g., `PaymentForm`). Afterwards, collect the same consents listed above and display a confirmation message:

- “Thank you for your payment. Your Holter kit will be shipped to you shortly. We will contact you via email and SMS with tracking information.”

Completion

All paths converge on a completion screen that summarises the user’s next steps. Use `Card` components with icons and clear headings. Include contact information for support.

Validation & Branching Rules

1. **Age requirement:** Users must be ≥ 18 years old (calculated from DOB) to proceed beyond Step 0.
2. **Contraindications:** Answering **Yes** to any contraindication stops the flow; users are advised to contact a GP or emergency services.
3. **Symptom presence:** At least one symptom must be selected for insured users to qualify for reimbursement. If no symptoms, the user is informed that screening is not reimbursed and may choose self-pay.
4. **Insurance status:** If the user lacks Swiss insurance, they follow the self-pay path by default.
5. **Insurance model:** Determines which branch of Stage 3 is presented (Standard/Flex → own GP or partner GP; HMO → own GP only; Telmed → hotline instructions).
6. **Consents:** All required checkboxes must be selected before submission (no partial acceptance).

Copy & Tone

- Use clear, plain language. Avoid medical jargon unless supplemented with tooltips.
- Highlight benefits (“You may be eligible for reimbursement”) in a positive, motivational tone.
- Provide concise explanations for why certain information is needed (e.g., “We ask about recent hospitalisations to ensure Holter monitoring is safe for you”).
- Emergency advice should be in a red `Alert` with bold text and an icon to draw attention.

Design Specifications

- **Colours & Typography:** Inherit primary and secondary colours from the site’s Tailwind configuration (e.g., `text-primary`, `bg-card`, `border-border`) ¹. Use the default sans font (IBM Plex Sans).
- **Spacing:** Use consistent `space-y-4` between form sections and `space-y-2` between fields.
- **Cards & Containers:** Wrap each step’s content in a `Card` component with a `CardHeader` (title) and `CardContent` (fields). Use `CardFooter` for navigation buttons (“Back”, “Continue”).
- **Buttons:** Use `Button` with variant `primary` for progression, `secondary` for back navigation, and `destructive` for cancel/exit actions.

- **Progress Indicator:** Position a horizontal stepper at the top of the form on desktop and convert to a vertical sidebar on mobile if space is limited. Steps should be labelled and highlight the current step.
- **Modals & Alerts:** Use `Dialog` for emergency follow-ups and `Alert` for eligibility and error messages. Ensure modals cannot be dismissed without acknowledging the message when high-risk conditions are present.
- **Accessibility:** Label all inputs, ensure sufficient colour contrast and support keyboard navigation. Use `aria-live` regions for dynamic messages (e.g., eligibility reward).

Integration Notes

- **Framework:** The site is built with React and Next.js; leverage shadcn/ui components. Manage form state using React context or a state machine (e.g., XState) to handle branching logic and maintain data across steps.
- **Backend:** Use Supabase or existing backend to send OTP codes, store partial and final submissions, and generate referral packet PDFs. Payment integration can use Stripe or Square; handle payment events server-side.
- **Routing:** The CTA on <https://smartholter.netlify.app/> should link to `/eligibility` or similar, which loads the multistep form. Use client-side routing to avoid page reloads.
- **Analytics:** Instrument each step with events (e.g., `eligibility_step_completed`, `contraindication_flagged`) to measure drop-off. Consider A/B testing copy or step order.

This specification can be used by designers and developers to implement the new questionnaire within the existing site, ensuring consistency with the design system and providing a user-friendly, eligibility-first flow.

¹ `tailwind.config.ts`

<https://github.com/yangsi7/skinn-convert-swiss-pages/blob/42fd4109855e45af44693a32c48f651f3af80936/tailwind.config.ts>